PRAHARI

প্রহরী · The Sentinel

An AI-Powered National Strategic Intelligence & Defense Readiness Platform for Bangladesh

DRISHTI Geopolitical & Diplomatic Intelligence	**RAKKHOK** Defense Asset Lifecycle & Readiness Monitoring	**SHOMUDRO** Maritime Domain Awareness & Dark Vessel Detection

AI HACKATHON 2026 · PROJECT BLUEPRINT & FULL FEATURE PLAN

Category: AI for National Impact · Defense, Security & Governance

Table of Contents

1. Executive Summary
2. The Problem: Why Bangladesh Needs PRAHARI
3. Solution Overview — One Platform, Three Shields
4. Module 1: DRISHTI — Geopolitical & Diplomatic Intelligence Engine
5. Module 2: RAKKHOK — Defense Asset Lifecycle & Readiness Monitor
6. Module 3: SHOMUDRO — Maritime Domain Awareness & Dark Vessel Detection
7. Cross-Cutting Platform Features
8. System Architecture
9. AI / ML Technical Approach
10. Data Sources & Feasibility
11. Hackathon MVP — What We Will Demo
12. Implementation Roadmap
13. Security, Ethics & Governance
14. Impact, Risks & Mitigation
15. Conclusion

1. Executive Summary

Bangladesh sits at one of the most strategically contested crossroads in Asia. Great-power competition in the Bay of Bengal, shifting trade blocs, sensitive relationships with neighbours, and a maritime zone larger than the country's landmass all demand decisions that are fast, evidence-based, and centred exclusively on Bangladesh's national interest. Today, the information needed for those decisions exists — in news wires, treaties, satellite feeds, shipping signals, and maintenance logbooks — but it is scattered, unstructured, and analysed manually, slowly, and inconsistently.

PRAHARI (প্রহরী, **'The Sentinel'**) is a sovereign, AI-powered decision-support platform that converts this ocean of open and internal data into one clear question answered daily: **"What is best for Bangladesh — and what must Bangladesh avoid?"** It is built as three integrated shields:

- **DRISHTI** (দৃষ্টি — Vision): a geopolitical and diplomatic intelligence engine that continuously reads global events, treaties, trade deals, and bilateral relationships between any countries in the world, then translates them into Bangladesh-specific policy recommendations, opportunity alerts, and risk warnings.
- **RAKKHOK** (রক্ষক — Guardian): a defense asset lifecycle and readiness monitor that tracks every aircraft, vessel, vehicle, radar, and spare part — flight hours, service life, expiry dates, maintenance due, fleet availability — and predicts failures before they happen.
- **SHOMUDRO** (সমুদ্র — Sea): a maritime domain awareness system for the Navy and Coast Guard that fuses satellite imagery, AIS ship signals, and radio-frequency data to detect "dark vessels" that switch off their transponders — the signature behaviour of smuggling, human trafficking, illegal fishing, and ship-to-ship contraband transfers in the Bay of Bengal.

PRAHARI is deliberately designed to be **buildable**: every module runs on data that is publicly or commercially available today (GDELT, UN treaty databases, Sentinel-1 SAR imagery, AIS feeds, Global Fishing Watch), with a clear path to integrating classified government feeds later. The hackathon MVP demonstrates all three shields end-to-end on real Bay of Bengal data.

"Data is the new territory. PRAHARI makes sure Bangladesh never cedes an inch of it."

2. The Problem: Why Bangladesh Needs PRAHARI

2.1 The strategic intelligence gap

- Global events that directly affect Bangladesh — a port deal between two other nations, a sanctions regime, a new trade corridor, a border agreement between neighbours — are analysed reactively, often weeks after the window to act has closed.
- Analysis is fragmented across ministries (Foreign Affairs, Commerce, Defence, Shipping) with no shared analytical picture and no institutional memory of how past decisions played out.
- There is no systematic ‘Bangladesh-first’ filter: reports describe what happened, not what Bangladesh should do, avoid, negotiate, or hedge against.

2.2 The readiness gap

- Defense assets — aircraft, patrol vessels, vehicles, radars — have service lives, overhaul schedules, and consumable expiry dates tracked largely on paper or isolated spreadsheets.
- The result: surprise groundings, unavailable platforms during emergencies (cyclone response, search & rescue), costly emergency procurement, and no fleet-level view of “how ready are we, right now?”

2.3 The maritime blind spot

- Bangladesh's maritime area (~118,813 km^2 of territorial waters and EEZ) is vast relative to available patrol assets. Physical patrol alone cannot cover it.
- Vessels engaged in drug smuggling, human trafficking, and illegal, unreported and unregulated (IUU) fishing routinely **switch off their AIS transponders** (“going dark”) or perform mid-sea ship-to-ship transfers far from the coast.
- Illegal fishing alone drains an estimated hundreds of millions of dollars in marine resources annually, while trafficking routes through the Bay of Bengal remain among the most active in the region.

Each gap is individually serious. Together they share one root cause — **data exists, intelligence does not**. PRAHARI closes that gap with a single sovereign AI platform.

3. Solution Overview — One Platform, Three Shields

PRAHARI is a modular platform: a shared data-and-AI core with three mission modules on top. Every module feeds a single **National Situation Dashboard** with role-based views for the Prime Minister's Office, Ministry of Foreign Affairs, Armed Forces Division, Navy, Coast Guard, and analysts.

Module	Function	Primary Users	Core Question Answered
DRISHTI	Geopolitical & diplomatic intelligence	MoFA, PMO, Commerce, AFD	“What should Bangladesh do about event X?”
RAKKHOK	Asset lifecycle & predictive readiness	Army, Navy, Air Force logistics	“Which assets are at risk, expiring, or due service?”
SHOMUDRO	Maritime surveillance & dark-vessel detection	Navy, Coast Guard, Fisheries	“Which ships are hiding, and where?”

3.1 Design principles

- **Bangladesh-first analytics:** every output ends in a recommendation scored for national interest — economic, security, diplomatic — never just a summary.
- **Human-in-the-loop:** PRAHARI advises; authorized officers decide. Every recommendation carries its evidence chain and confidence score.
- **Sovereign & air-gappable:** deployable on government infrastructure; open-weight models fine-tuned locally so no sensitive query ever leaves the country.
- **Open-data-first feasibility:** the entire MVP runs on public data; classified feeds are a plug-in, not a prerequisite.
- **Bilingual by design:** full Bangla + English interface, reports, and voice briefings.

4. Module 1: DRISHTI — Geopolitical & Diplomatic Intelligence Engine

DRISHTI continuously monitors the world's political, diplomatic, economic, and military activity and answers one question for every development, anywhere on the planet: **what does this mean for Bangladesh, and what should Bangladesh do (or deliberately not do)?**

4.1 How it works — the intelligence pipeline

- **Ingest:** 24/7 collection from global news (GDELT: 100+ languages, updated every 15 minutes), government press releases, UN/WTO/treaty databases, sanctions lists, trade statistics (UN Comtrade), think-tank reports, parliamentary records, and social/official media of key states.
- **Extract:** LLM-based information extraction converts raw text into structured events — who did what with whom (actors, action type, sector, magnitude, location, date) — in any language, normalised into a single event schema.
- **Connect:** events populate a living **Geopolitical Knowledge Graph**: nations, leaders, ministries, companies, ports, treaties, deals, disputes — and the evolving relationships between them (ally, creditor, rival, supplier, guarantor).
- **Reason:** a Bangladesh-tuned analysis layer scores each development on relevance to Bangladesh, estimates first/second/third-order effects (trade, remittance, security, energy, water, labour markets), and drafts options.
- **Recommend:** outputs a structured brief: situation, why it matters to Bangladesh, DO recommendations, AVOID recommendations, hedging options, timeline of decision windows, confidence, and full source citations.

4.2 Feature list — DRISHTI

#	Feature	Description
D1	Global Event Radar	Real-time feed of world events auto-ranked by ‘Bangladesh Relevance Score’ (0–100), filterable by region, sector, and severity.
D2	Bilateral Relationship Tracker	Live health index for every Bangladesh bilateral relationship (and key third-party pairs, e.g. two partner nations dealing with each other), tracking trend, drivers, open disputes, and pending agreements.
D3	Treaty & Deal Analyzer	Upload or auto-fetch any treaty/MoU/contract text; AI extracts obligations, exit clauses, penalty terms, asymmetries, and compares against similar deals other countries signed — flagging clauses unfavourable to Bangladesh.
D4	Do / Avoid Advisor	For any event or proposed decision, generates a two-column national-interest brief: recommended actions vs. actions to avoid, each with reasoning, precedent cases, and confidence.
D5	Scenario Simulator (war-gaming)	‘What-if’ engine: simulate outcomes of policy options (join corridor X? sign deal Y? stay neutral in dispute Z?) across economic, security, and diplomatic dimensions using historical analogues + causal models.
D6	Early-Warning Alerts	Push alerts when leading indicators cross thresholds: escalation near borders, commodity shocks affecting imports, policy shifts by major garment buyers, sanctions exposure of partners.
D7	Negotiation Copilot	Pre-meeting dossier for any counterpart country/delegation: their interests, red lines, recent positions, internal pressures, and suggested leverage points for Bangladesh.

#	Feature	Description
D8	Daily National Brief	Auto-generated morning intelligence brief (Bangla + English, text + audio) tailored per role: PMO, MoFA, AFD, Commerce.
D9	Institutional Memory	Searchable archive linking past decisions to observed outcomes, so future recommendations learn from Bangladesh's own history.
D10	Disinformation Watch	Detects coordinated narrative campaigns targeting Bangladesh's interests, image, or economy across media, and traces origin clusters.

5. Module 2: RAKKHOK — Defense Asset Lifecycle & Readiness Monitor

RAKKHOK is the ‘guardian of the guardians’: a single digital registry and predictive-maintenance brain for every defense asset Bangladesh operates — aircraft, ships, vehicles, radars, generators, weapons platforms, and their spare-part inventories. It answers, at any moment: **what do we have, what condition is it in, what is expiring, what will fail next, and how ready are we?**

5.1 How it works

- **Digital Asset Registry:** every asset gets a digital record (a lightweight ‘digital twin’): type, manufacturer, induction date, design service life, usage counters (flight hours, engine hours, nautical miles, rounds fired), overhaul history, and current status.
- **Usage & sensor ingestion:** data enters via existing logbooks (digitised with OCR), maintenance management exports, and — where available — IoT sensors (vibration, temperature, oil quality) retrofitted on high-value platforms.
- **Predictive models:** survival-analysis and gradient-boosted models estimate remaining useful life (RUL) per component; anomaly detection on sensor streams flags degradation between scheduled services.
- **Readiness engine:** rolls individual asset states up into unit-, base-, and force-level readiness percentages, and simulates availability for scenarios (cyclone response, extended patrol surge).

5.2 Feature list — RAKKHOK

#	Feature	Description
R1	Fleet Health Dashboard	One screen per force: total assets, operational / in-maintenance / grounded counts, readiness %, trend lines, and drill-down to individual tail/hull numbers.
R2	Service-Life & Expiry Tracker	Automatic countdowns for airframe life, engine TBO, hull surveys, battery/consumable expiry, certification renewals — with 180/90/30-day escalating alerts. Directly answers: ‘Is this aircraft expired? Is it still being flown? How many are actually flyable?’
R3	Predictive Maintenance (RUL)	ML-predicted remaining useful life per critical component; ranks the fleet by failure risk so maintenance budgets go where the risk is.
R4	Spare-Parts Intelligence	Inventory levels vs. predicted demand; flags parts that will run out before the next procurement cycle; suggests cannibalisation trade-offs explicitly instead of ad-hoc.
R5	Maintenance Scheduler	Auto-drafts optimal maintenance calendars that maximise fleet availability (never lets too many platforms be in the shop at once).
R6	Mission Readiness Simulator	‘If we surge 6 patrol vessels for 21 days, what breaks and when?’ — scenario stress-testing of the fleet.
R7	Lifecycle Cost Analytics	Total cost of ownership per platform; flags assets whose upkeep now exceeds replacement economics — evidence for procurement decisions.
R8	Incident & Defect Registry	Structured defect reporting with photo/voice input from technicians (Bangla supported); AI clusters recurring defects across the fleet to spot systemic issues.
R9	Procurement Advisor (links to DRISHTI)	When replacement is needed, cross-references DRISHTI's supplier-country risk analysis: sanctions exposure, spare-part dependency, geopolitical reliability of the vendor nation.

6. Module 3: SHOMUDRO — Maritime Domain Awareness & Dark Vessel Detection

SHOMUDRO gives the Navy and Coast Guard persistent, satellite-scale eyes over the entire Bay of Bengal EEZ — including vessels that **do not want to be seen**. Its core capability is detecting ‘dark vessels’: ships that switch off their AIS transponders to hide smuggling, human trafficking, illegal fishing, or mid-sea ship-to-ship (STS) transfers.

6.1 How dark-vessel detection works (the core innovation)

- **Layer 1 — AIS baseline:** ingest live AIS feeds (every legal vessel broadcasts identity, position, course, speed). Build normal-behaviour models per vessel type, route, and season for the Bay of Bengal.
- **Layer 2 — Satellite SAR imagery:** ESA Sentinel-1 synthetic-aperture radar images the sea surface day/night, through cloud and monsoon. Computer-vision models detect every metal vessel in each scene as a bright radar return.
- **Layer 3 — Cross-correlation (the ‘dark’ test):** every SAR-detected vessel is matched against AIS. **A ship visible on radar but absent from AIS = dark vessel.** Each gets a risk score from context: location (near known trafficking corridors?), time (night?), behaviour, and history.
- **Layer 4 — Behaviour analytics on AIS itself:** even before going dark, suspicious ships betray themselves: sudden AIS gaps, loitering in open sea, two vessels converging to near-zero distance and matching speed (the classic STS-transfer signature), impossible speed jumps (GPS/AIS spoofing), or identity mismatches. Sequence models flag these patterns in real time.
- **Layer 5 — Optional RF augmentation:** commercial RF-geolocation constellations (e.g. radar/comms emissions mapping) can be added later to geolocate emitting-but-dark vessels between satellite passes.
- **Layer 6 — Interdiction support:** confirmed high-risk contacts generate an actionable packet for the nearest patrol asset: last fix, predicted drift/track (from current and wind data), intercept vector, and evidence log for prosecution.

6.2 Feature list — SHOMUDRO

#	Feature	Description
S1	Live Maritime Picture	Unified map of the entire Bangladesh EEZ: all AIS tracks, SAR detections, dark-vessel flags, patrol asset positions, and weather overlay.
S2	Dark Vessel Detector	SAR-vs-AIS correlation engine producing a ranked daily list of unidentified vessels with position, size estimate, heading, and risk score.
S3	STS Rendezvous Alert	Detects two-vessel mid-sea meetings (drug hand-offs, fuel/fish transshipment, trafficking transfers) from track convergence patterns — including when one party is dark.
S4	AIS Spoofing & Identity Check	Flags cloned MMSIs, impossible tracks, and vessels whose broadcast identity conflicts with their radar size class.
S5	IUU Fishing Monitor	Fishing-behaviour classification (trawling/loitering signatures) inside restricted zones, marine sanctuaries, and the 65-day ban period; foreign intrusion alerts at EEZ boundaries.
S6	Trafficking Corridor Watch	Persistent monitoring of known human-smuggling departure zones and seasonal route models; night-movement anomaly alerts for small-craft clusters.
S7	Patrol Optimizer	Recommends daily patrol routes that maximise coverage of current risk hotspots given fuel and asset constraints — multiplying the effect of a limited fleet.

#	Feature	Description
S8	Port & Anchorage Watch	Monitors Chattogram/Mongla/Payra anchorages for unusual dwell times, undeclared arrivals, and sanctioned-vessel port calls.
S9	Evidence Package Generator	One-click legal evidence bundle per incident: imagery, tracks, timestamps, chain-of-custody hashes — built for court admissibility.
S10	Environmental Sentinel	Oil-spill and slick detection from the same SAR feed; cyclone-time vessel safety monitoring and SAR (search & rescue) support.

7. Cross-Cutting Platform Features

#	Feature	Description
C1	National Situation Dashboard	Single command view combining all three modules; role-based access (PMO / MoFA / AFD / Navy / Coast Guard / analyst).
C2	Ask-PRAHARI (Bangla + English)	Natural-language interface over the whole knowledge graph: 'Which of our patrol vessels can reach grid X in 6 hours?', 'Summarise our obligations under treaty Y.'
C3	Explainable AI layer	Every score, alert, and recommendation exposes its evidence chain, sources, and confidence — no black-box advice.
C4	Audit & accountability log	Immutable log of who saw what and which recommendations informed which decisions.
C5	Offline / air-gapped mode	Full functionality on sovereign infrastructure with periodic batch data sync; no dependency on foreign cloud.
C6	Voice briefings & mobile app	Daily audio brief and secure mobile companion for senior decision-makers.
C7	API & inter-agency bus	Standards-based APIs so ministries and forces integrate without replacing existing systems.

8. System Architecture

8.1 Layered design

Layer	Role	Key Components
1. Ingestion	Connectors & schedulers	GDELT stream, RSS/news APIs, treaty DB scrapers, UN Comtrade, AIS feed (aisstream.io / Spire), Sentinel-1 via Copernicus Data Space, Global Fishing Watch API, logbook OCR uploads, IoT gateways (MQTT).
2. Data platform	Lake + warehouse + graph	Object store (raw imagery/text) → PostgreSQL + PostGIS (geospatial), TimescaleDB (sensor/AIS time-series), Neo4j (geopolitical knowledge graph), vector DB (Qdrant) for semantic search.
3. AI layer	Models & pipelines	LLM services (fine-tuned open-weight models, e.g. Llama-class, hosted locally), CV pipeline for SAR (YOLO-family detector), anomaly/sequence models (Transformer on AIS tracks), survival models for RUL, simulation engine.
4. Application	Services & UX	FastAPI micro-services, Kafka event bus, React + MapLibre dashboard, mobile app, alerting (SMS/secure push), report generator (Bangla PDF engine).
5. Security	Zero-trust envelope	RBAC + MFA, end-to-end encryption, network segmentation per classification level, full audit logging, deployable air-gapped.

8.2 Data flow (one sentence per shield)

- **DRISHTI:** world text → event extraction → knowledge graph → Bangladesh-relevance scoring → recommendation brief.
- **RAKKHOK:** logbooks/sensors → digital twin update → RUL & readiness models → alerts + maintenance schedule.
- **SHOMUDRO:** AIS + SAR scene → vessel detection → AIS correlation → dark/anomaly scoring → interdiction packet.

9. AI / ML Technical Approach

Task	Approach	Evaluation
Event extraction from global text	Fine-tuned multilingual LLM with structured-output (JSON schema) decoding; GDELT event codes as weak supervision.	Precision/recall vs. hand-labelled event set; ≥ 85% F1 target.
Bangladesh Relevance Scoring	Hybrid: graph proximity (how connected is the event to BD nodes) + LLM judgement + trade/exposure statistics.	Ranking agreement with expert analysts (nDCG).
Do/Avoid recommendation	Retrieval-augmented generation over knowledge graph + precedent archive; chain-of-analysis prompting; self-consistency checks; mandatory citations.	Expert review panel scoring; citation validity rate.
Scenario simulation	Causal graphs + Monte-Carlo over historical analogue outcomes; sensitivity analysis surfaced to the user.	Backtesting on 20 historical BD-relevant events.
SAR vessel detection	YOLOv8/11 trained on open SAR ship datasets (SSDD, LS-SSDD, xView3 — xView3 is purpose-built for dark-vessel detection), fine-tuned on Bay of Bengal scenes.	xView3 metrics: detection F1, close-to-shore F1.
Dark matching (SAR↔AIS)	Spatio-temporal association with track interpolation to satellite-pass time; Hungarian matching with gating.	Match accuracy on vessels with known ground truth.
AIS anomaly / STS detection	Sequence Transformer + rule layer (gap detection, rendezvous geometry: distance < 200 m, speed ≈ 0, duration > 30 min).	Alert precision on Global Fishing Watch labelled encounters.
Remaining Useful Life (RUL)	Gradient boosting + Weibull survival analysis on usage counters; LSTM on sensor streams where IoT exists (NASA C-MAPSS as pre-training benchmark).	RUL error (RMSE) and alert lead-time.
Bangla NLP & briefing	Bangla-capable LLM fine-tune for report generation and voice (TTS) briefings.	Fluency ratings by native reviewers.

10. Data Sources & Feasibility

The decisive feasibility argument: **every core capability is demonstrable with data that is free or low-cost today.**

Source	What it provides	Cost	Feeds
GDELT Project	Global events, 100+ languages, 15-min updates	Free	DRISHTI event radar
UN Treaty Collection / WTO / national gazettes	Treaties, trade agreements, legal texts	Free	Treaty & Deal Analyzer
UN Comtrade, World Bank, IMF	Trade, debt, macro exposure data	Free	Relevance & impact scoring
Sentinel-1 (Copernicus Data Space)	SAR imagery of Bay of Bengal, ~6–12 day revisit, all-weather	Free	Dark vessel detection
xView3-SAR dataset	1,000+ labelled SAR scenes built specifically for dark-vessel ML	Free	Model training
AIS feeds (aisstream.io free tier; Spire/exactEarth commercial)	Live global ship positions	Free tier → paid scale-up	Maritime picture, anomaly detection
Global Fishing Watch API	Fishing activity, encounters (STS), loitering, AIS-gap events	Free	IUU monitor, STS validation
Open-Meteo / BMD	Wind, current, cyclone data	Free	Drift prediction, patrol planning
Government feeds (later phase)	Logbooks, MMS exports, coastal radar, classified reporting	Internal	RAKKHOK + fusion upgrade

Higher-frequency commercial layers (daily SAR tasking from ICEYE/Umbra, RF geolocation from HawkEye 360-class providers) are a scale-up path, not an MVP requirement.

11. Hackathon MVP — What We Will Demo

A judge-ready, end-to-end demonstration on **real Bay of Bengal data**, built in the hackathon window:

Demo storyline (7 minutes)

- **Scene 1 — DRISHTI (2 min):** live event feed ranked by Bangladesh Relevance; click one real recent event → auto-generated Do/Avoid brief with cited sources; ask a question in Bangla and get a spoken answer.
- **Scene 2 — RAKKHOK (2 min):** demo fleet of 25 synthetic-but-realistic assets; dashboard shows readiness %, an aircraft crossing its service-life threshold triggers an alert; RUL ranking shows which platform to service first.
- **Scene 3 — SHOMUDRO (3 min):** a real Sentinel-1 scene of the Bay of Bengal processed live: model detects vessels → overlays AIS → highlights unmatched dark contacts on the map → one contact shows an STS-rendezvous pattern → system generates the interdiction packet. This is the wow moment.

MVP build scope

Scope	Items
Included	GDELT ingestion + relevance ranking; RAG-based Do/Avoid brief; asset registry + expiry alerts + simple RUL model; Sentinel-1 detection (pre-trained on xView3, one AOI); AIS overlay + dark matching; STS rule detector; unified map dashboard; Bangla output.
Excluded (roadmap)	Classified integrations, IoT retrofits, RF constellation data, full scenario simulator, mobile app, air-gap hardening.

12. Implementation Roadmap

Phase	Timeline	Work	Exit criteria
Phase 0 — Hackathon	48 hours	MVP as above; open data only; single AOI in the Bay of Bengal.	Working demo, pitch, this blueprint.
Phase 1 — Pilot	Months 1–6	Harden pipelines; daily Sentinel-1 coverage of full EEZ; Coast Guard pilot cell (5 analysts); RAKKHOK pilot with one squadron/flotilla's real logbooks; feedback loops.	Pilot report; detection precision ≥ 80%; 2 real interdictions supported.
Phase 2 — National deployment	Months 7–18	Sovereign data centre deployment; MoFA + AFD onboarding of DRISHTI; commercial AIS + tasked SAR; IoT retrofit on 3 high-value platforms; institutional-memory archive seeded.	Inter-agency operating picture; readiness reporting live.
Phase 3 — Full sentinel	Months 19–36	RF-geolocation layer; scenario simulator v2; regional data-sharing (IORA/BIMSTEC partners) on IUU fishing; continuous model retraining unit.	Persistent EEZ coverage; measurable smuggling-route disruption.

13. Security, Ethics & Governance

- **Human decision authority:** PRAHARI never acts autonomously; it informs interdiction, diplomacy, and maintenance decisions made by authorised officers under existing law and rules of engagement.
- **Legal alignment:** maritime operations follow UNCLOS and national law; evidence packages are designed for judicial process, not extrajudicial action.
- **Privacy proportionality:** the system tracks vessels and state-level actors — not private citizens; disinformation analysis works at the narrative/cluster level, not individual surveillance.
- **Explainability & auditability:** every recommendation carries sources and confidence; an immutable audit trail supports parliamentary and internal oversight.
- **Sovereignty of data & models:** open-weight models fine-tuned and hosted in-country; no query, document, or asset record leaves national infrastructure.
- **Bias & error control:** red-team reviews of recommendations, dual-analyst confirmation for high-severity alerts, and published error-rate reporting to prevent automation over-trust.

14. Impact, Risks & Mitigation

14.1 Expected impact

- **Diplomatic:** decision windows caught in hours instead of weeks; negotiations entered with machine-prepared, precedent-backed dossiers.
- **Economic:** recovered fisheries revenue from IUU deterrence; avoided losses from unfavourable contract clauses; maintenance savings from predictive scheduling (industry benchmarks: 10–25% maintenance cost reduction).
- **Security & humanitarian:** disruption of trafficking and smuggling routes; faster search-and-rescue; higher fleet availability during cyclones.

14.2 Key risks

Risk	Mitigation
Satellite revisit gaps (free tier)	Fuse AIS-anomaly detection (continuous) with SAR (periodic); commercial tasking in Phase 2.
Model hallucination in briefs	Strict RAG with mandatory citations; refusal on low evidence; analyst sign-off before circulation.
Data-sharing resistance between agencies	Start with one champion agency (Coast Guard); federated architecture — agencies keep custody of their data.
Adversary adaptation (spoofing, small wooden craft)	Multi-sensor fusion roadmap (optical, RF); wooden-craft detection via optical imagery in Phase 3.
Talent retention	University partnership pipeline (EWU/BUET); documented, containerised stack to reduce key-person risk.

15. Conclusion

PRAHARI is three shields on one platform: **DRISHTI** watches the world and tells Bangladesh what to do and what to avoid; **RAKKHOK** watches our own assets so nothing expires, fails, or surprises us; **SHOMUDRO** watches the sea and drags dark vessels into the light. Each module is independently valuable; together they form a sovereign nervous system for national decision-making.

Nothing here waits on science fiction. The data is flowing today, the models are proven in open benchmarks like xView3, and the MVP demonstrates the full loop — from a satellite image over the Bay of Bengal to an actionable interdiction packet, and from a headline anywhere on Earth to a Bangla-language recommendation on a decision-maker's phone. What has been missing is not technology, but the will to assemble it **for Bangladesh, by Bangladesh**.

PRAHARI — প্রহরী — because a nation that sees first, decides best.